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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

22GEX03 – AI for Engineers R2022

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Bonafide Certificate

Name _____

Programme **B.E.**, Branch **EEE**, Section _____

Semester **VII**, Roll No. / Reg. No. _____

Certified that this is a Bonafide record of work done by the above student
of the **22GEX03 – AI for Engineers** during the year **2026 - 27**.

Signature

Laboratory In- Charge

Signature

Head of the Department

Submitted for the End Semester Practical Examination held on

_____.

Examiner 1

Examiner 2

22GEX03 – AI FOR ENGINEERS

Programme & Branch	B.E. & Mechatronics Engineering	Sem	Category	L	T	P	SL*	Total	Credit
Prerequisites	Nil	7	OE	30	0	30	30	90	3

Preamble This course introduces software engineering concepts and agile principles which is to be acquired by software engineers and developers. It also focuses on providing hands-on experience in designing and developing software systems.

LIST OF EXPERIMENTS / EXERCISES:

1. Implementation of Simple and Multiple Linear Regression models for a sample dataset
2. Building and testing Naïve Bayes and Decision tree classification model for a sample dataset
3. Perform K-means clustering for a sample dataset
4. Perform image classification using CNN
5. Implementation of RNN for sequential data
6. Perform object detection using YOLO
7. Implement text summarization using a Large Language Model
8. Perform grammar correction and text rewriting using a Large Language Model
9. Develop domain-specific question answering system using a Large Language Model

Total: 30

REFERENCES/MANUAL/SOFTWARE:

1. Laboratory Manual

COURSE OUTCOMES:

On completion of the course, the students will be able to

		BT Mapped (Highest Level)
CO1	apply AI techniques such as classification, speech recognition, recommendation, conversational systems and vision-based models using suitable tools to solve real-world problems	Applying (K3)
CO2	apply regression, classification and clustering techniques to real-world datasets	Applying (K3) Precision(S3)
CO3	use deep learning architectures such as CNN, RNN and LSTM to solve image, sequence and object detection problems	Applying (K3) Precision(S3)
CO4	use generative AI tools for creating text, image and video content and apply them in simple engineering applications	Applying (K3) Precision(S3)
CO5	design and refine effective prompts to generate context-specific AI outputs	Applying (K3) Precision(S3)

Mapping of COs with POs and PSOs

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2
CO1	3	2	1		3								
CO2	3	2	1		3								
CO3	3	2	1		3								
CO4	3	2	1		3								
CO5	3	2	1		3								

1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy

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Expt. No.	Date	Title of the Experiment	COE (20)	Record (20)	Results (10)	Viva (10)	Total (60)	In-Charge Signature
1		Implementation of Simple and Multiple Linear Regression models for a sample dataset						
2		Building and testing Naïve Bayes and Decision tree classification model for a sample dataset						
3		Perform K-means clustering for a sample dataset						
4		Perform image classification using CNN						
5		Implementation of RNN for sequential data						
6		Perform object detection using YOLO						
7		Implement text summarization using a Large Language Model						
8		Perform grammar correction and text rewriting using a Large Language Model						
9		Develop domain-specific question answering system using a Large Language Model						
Average								

Completed Date:

Signature

Laboratory In-Charge

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Academic Year : **2026 - 27**
Course Code : **22GEX03**

Semester : **VII**
Year : **VI**

Course Name : **AI for Engineers**

Section : **A / B**

Rubrics for Continuous Assessment – Practical

Parameters	Scale	Marks	≥ 90%	89% ≥ 50%	< 50%
Conduction of Experiment	Problem Understanding and Program Development	20	Problem correctly understood; appropriate algorithm/logic designed and implemented independently without errors	Minor errors in problem understanding or logic design; requires limited guidance	Incorrect or incomplete logic; heavy dependence on instructor assistance; poor participation
Record	Code, Output & Explanation	20	Neat and well-commented code recorded; correct output/screenshots attached; explanation is error-free and well presented	Code or output has minor mistakes; explanation mostly correct with small errors	Incomplete or incorrect code recorded; improper output; explanation errors present
Results	Verification & Inference	10	Output/results clearly verified; inferences (e.g., accuracy, predictions, insights) are technically accurate and well-justified	Partial verification of results; inferences are acceptable but lack depth	Results not properly verified; incorrect or weak inference
Viva	Technical Knowledge in Programming Concepts	10	Demonstrates excellent conceptual clarity and confidently answers all questions	Basic understanding; answers most questions with minor difficulty	Limited understanding; unable to answer key questions
Model	Aim & Algorithm	10	Aim clearly stated; algorithm/approach correctly identified with proper steps	Aim or algorithm partially correct; minor omissions	Aim unclear; algorithm incorrect or incomplete
	Program Development /Implementation	40	Program developed and executed independently; systematic implementation; accurate output	Program developed with minor assistance; output mostly correct	Improper program logic or execution; incorrect or insufficient output
	Output, Analysis & Visualization	30	All outputs/metrics are correct, neat, and logically presented (including plots/visualizations where applicable)	Outputs/metrics mostly correct with minor errors in presentation	Major errors in output/metrics; poor presentation
	Results	10	Results accurately obtained and clearly interpreted	Results obtained but interpretation is partial	Results obtained but interpretation is weak or incorrect

	Viva	10	Excellent technical knowledge; clear and confident responses	Adequate knowledge; answers with hesitation	Insufficient technical understanding
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Expt. No. : 01	IMPLEMENTATION OF SIMPLE AND MULTIPLE LINEAR REGRESSION MODELS FOR A SAMPLE DATASET

AIM:

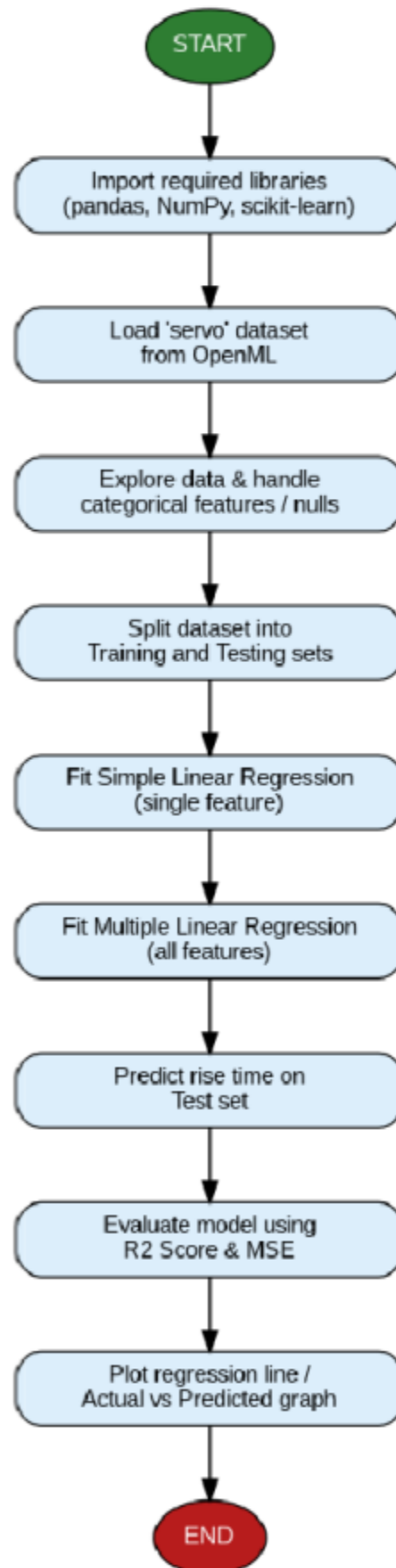
APPARATUS REQUIRED:

Personal Computer / Laptop with Windows 10 or above, Python 3.x, Jupyter Notebook / Google Colab, Python libraries - pandas, NumPy, scikit-learn, matplotlib, Internet connection to fetch dataset from OpenML (openml.org).

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 02	BUILDING AND TESTING NAIVE BAYES AND DECISION TREE CLASSIFICATION MODEL FOR A SAMPLE DATASET

AIM:

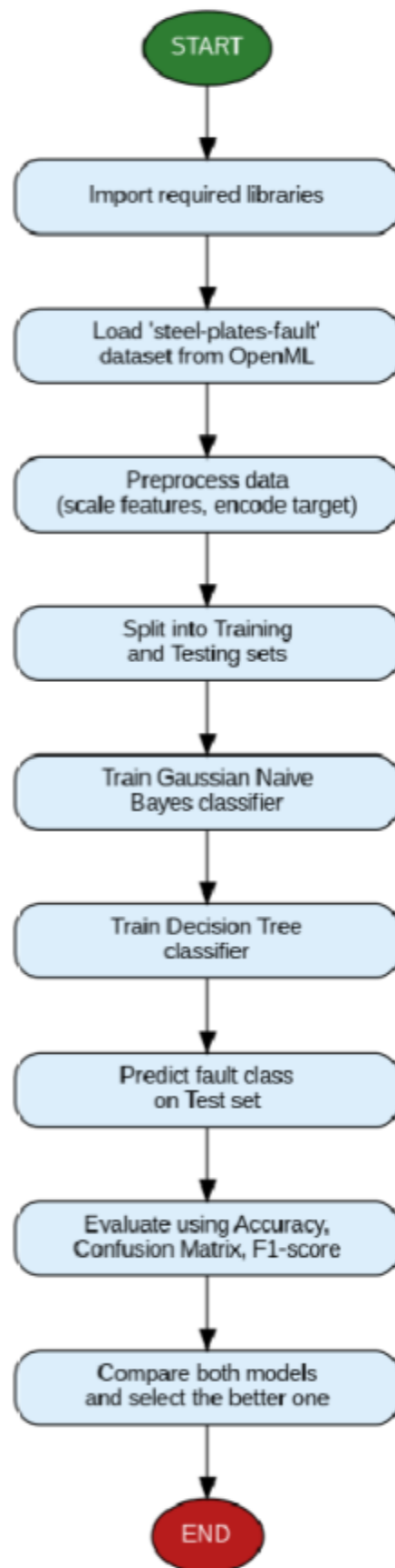
APPARATUS REQUIRED:

Personal Computer / Laptop, Python 3.x, Jupyter Notebook / Google Colab, Python libraries - pandas, NumPy, scikit-learn, matplotlib/seaborn, Internet connection to access OpenML.

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 03	PERFORM K-MEANS CLUSTERING FOR A SAMPLE DATASET

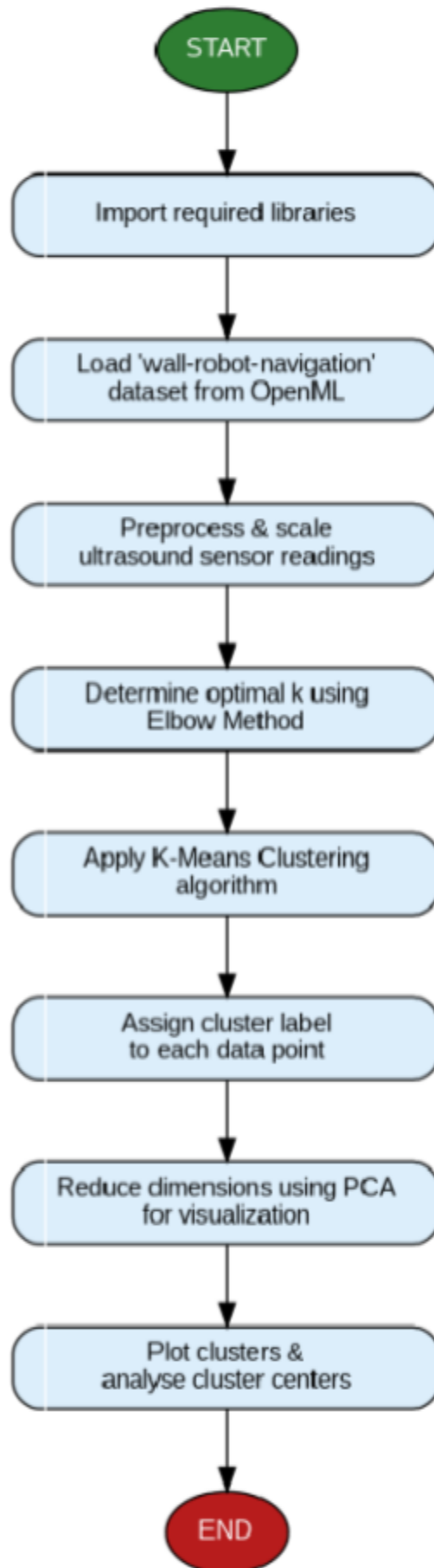
AIM:

APPARATUS REQUIRED:

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 04	PERFORM IMAGE CLASSIFICATION USING CNN

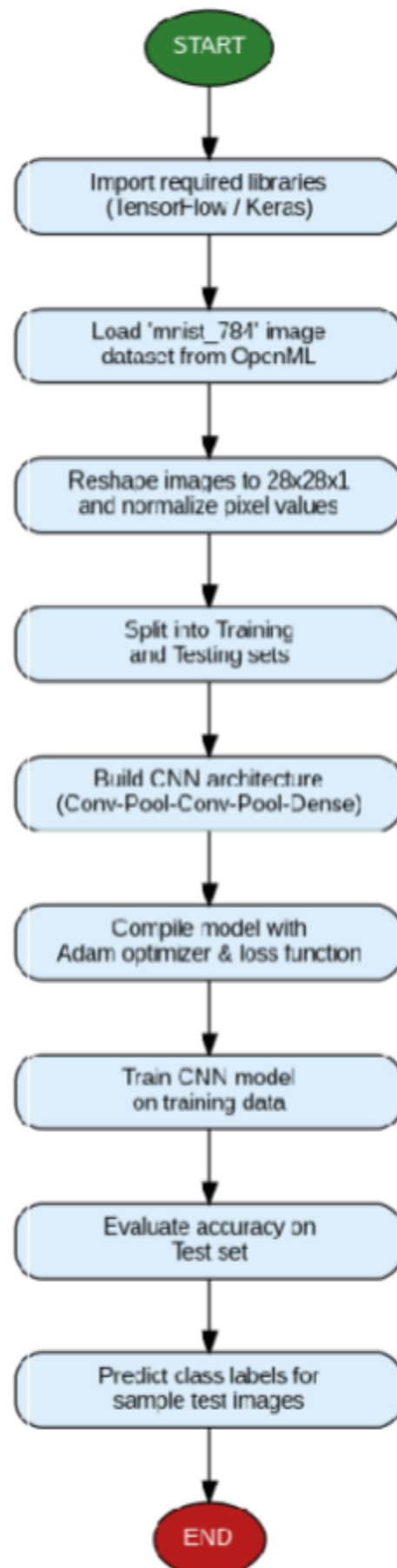
AIM:

APPARATUS REQUIRED:

Personal Computer / Laptop (GPU preferred), Python 3.x, Jupyter Notebook / Google Colab, Python libraries - TensorFlow/Keras, NumPy, scikit-learn, matplotlib, Internet connection to access OpenML.

THEORY:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 05	IMPLEMENTATION OF RNN FOR SEQUENTIAL DATA

AIM:

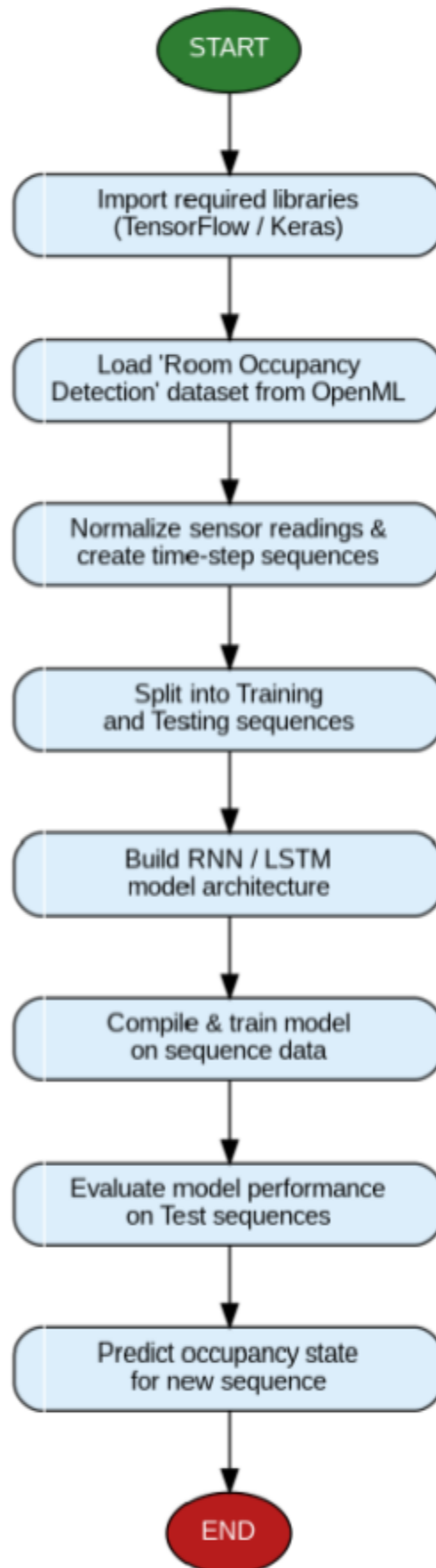
APPARATUS REQUIRED:

Personal Computer / Laptop, Python 3.x, Jupyter Notebook / Google Colab, Python libraries - TensorFlow/Keras, pandas, NumPy, scikit-learn, matplotlib, Internet connection to access OpenML.

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 06	PERFORM OBJECT DETECTION USING YOLO

AIM:

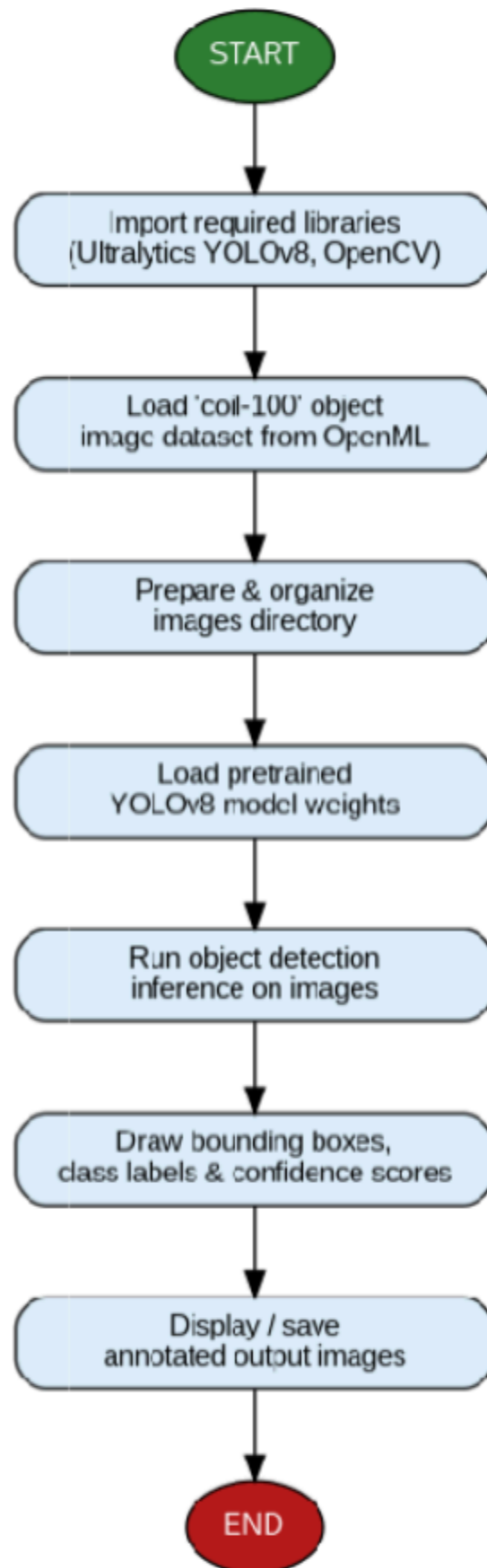
APPARATUS REQUIRED:

Personal Computer / Laptop (GPU preferred), Python 3.x, Jupyter Notebook / Google Colab, Python libraries - ultralytics (YOLOv8), OpenCV, matplotlib, Internet connection to access OpenML and download pretrained weights.

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 07	IMPLEMENT TEXT SUMMARIZATION USING A LARGE LANGUAGE MODEL

AIM:

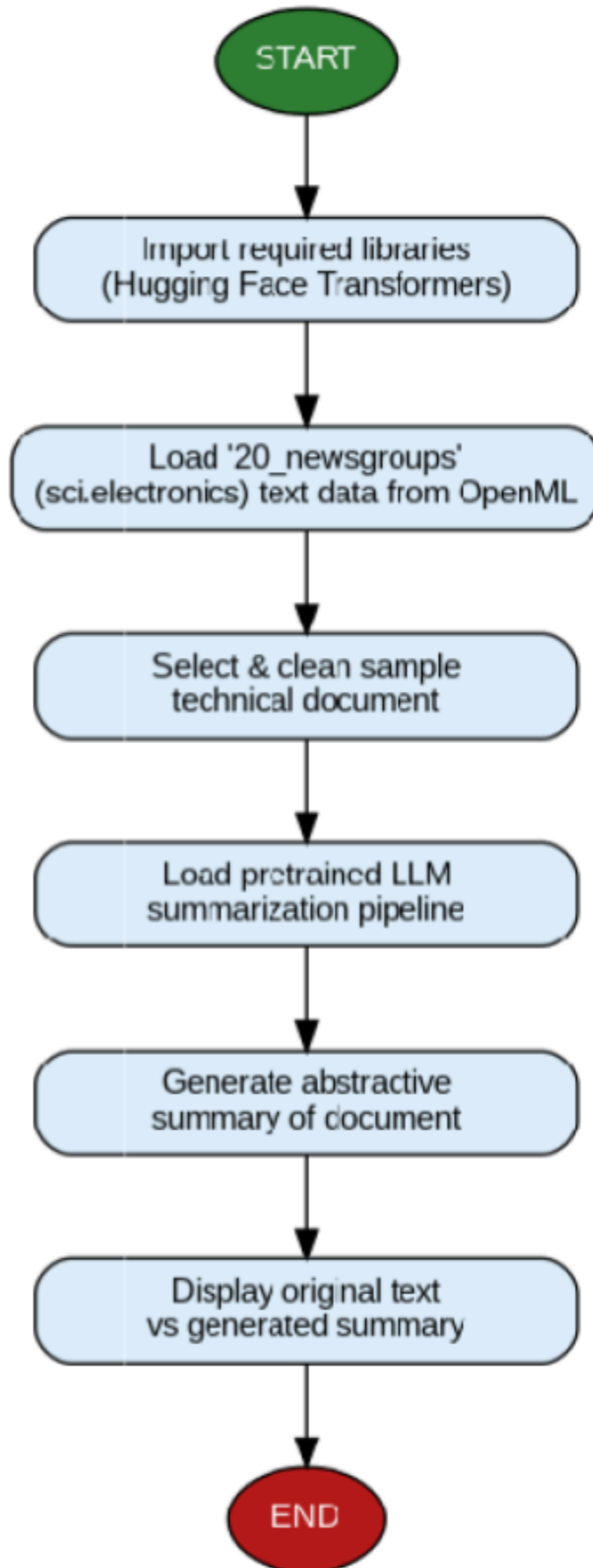
APPARATUS REQUIRED:

Personal Computer / Laptop, Python 3.x, Jupyter Notebook / Google Colab, Python library - Hugging Face transformers (pipeline API), Internet connection to access OpenML and download the pretrained LLM.

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 08	PERFORM GRAMMAR CORRECTION AND TEXT REWRITING USING A LARGE LANGUAGE MODEL

AIM:

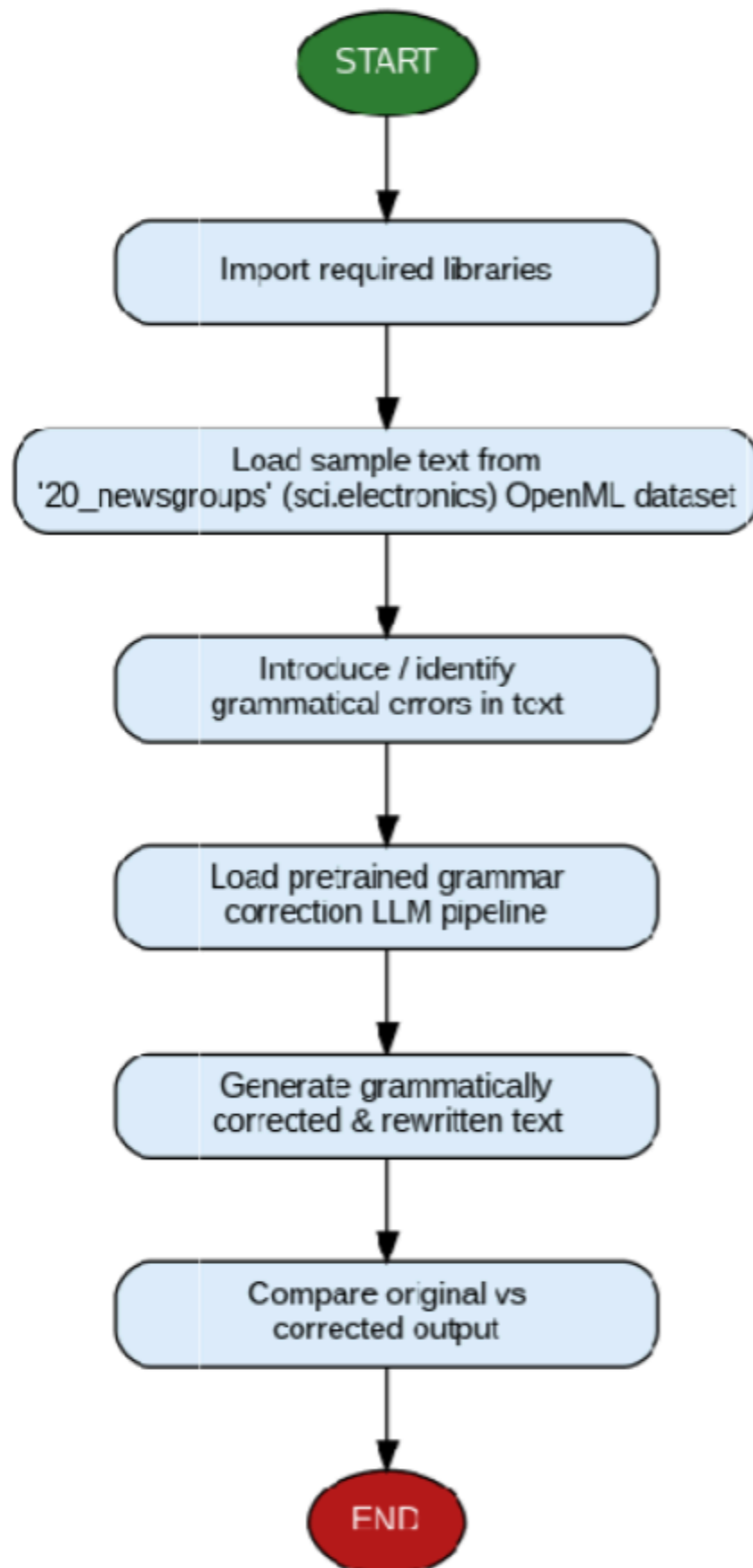
APPARATUS REQUIRED:

Personal Computer / Laptop, Python 3.x, Jupyter Notebook / Google Colab, Python library - Hugging Face transformers (pipeline API) with a grammar-correction model (e.g., "vennify/t5-base-grammar-correction"), Internet connection to access OpenML and the pretrained LLM.

THEORY:

ALGORITHM:

FLOW CHART:



SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

Expt. No. : 09	DEVELOP DOMAIN-SPECIFIC QUESTION ANSWERING SYSTEM USING A LARGE LANGUAGE MODEL

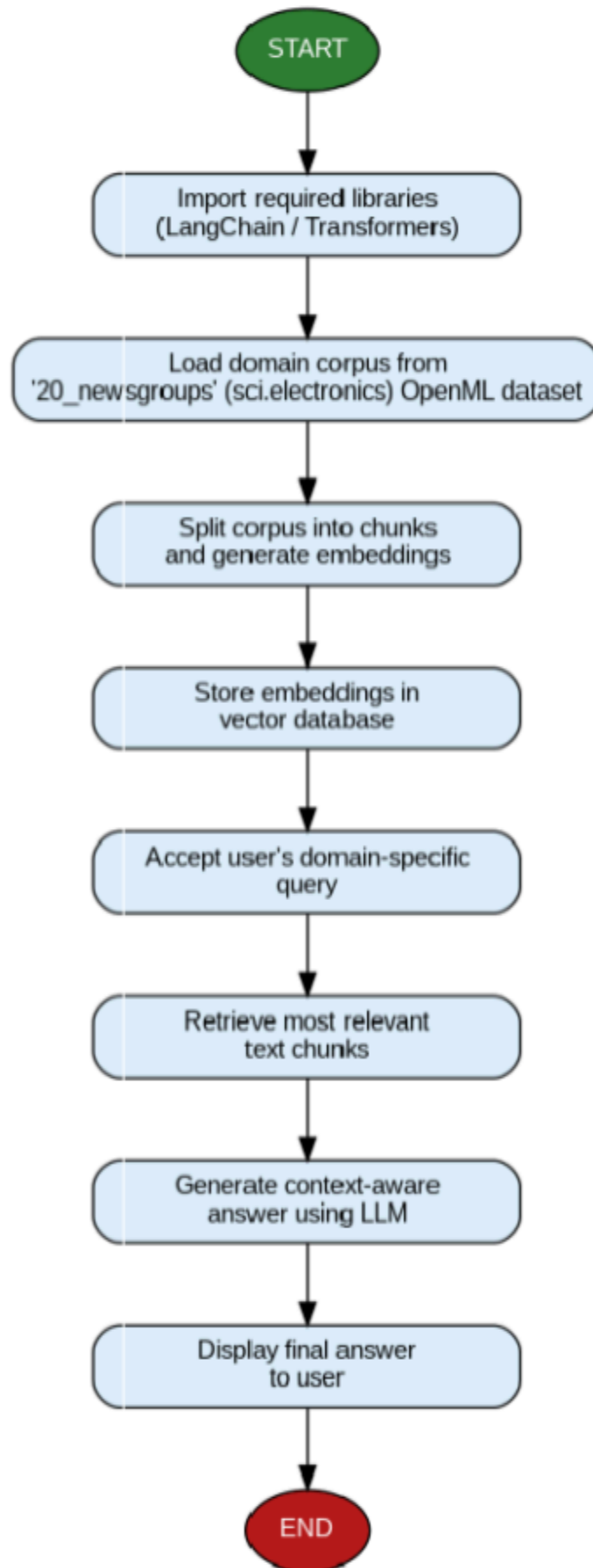
AIM:

APPARATUS REQUIRED:

Personal Computer / Laptop, Python 3.x, Jupyter Notebook / Google Colab, Python libraries - sentence-transformers, FAISS (vector store), Hugging Face transformers, Internet connection to access OpenML and the pretrained models.

THEORY:

FLOW CHART:



ALGORITHM:

SAMPLE PROGRAM:

SAMPLE OUTPUT:

S. No.	Parameters	Max. Marks	Marks Obtained
1.	Conduction of Experiment	20	
2.	Record	20	
3.	Results	10	
4.	Viva	10	
Total		60	

RESULT:

